Series convergence. Analyze the convergence of the following series:

$$(1)\sum_{n=1}^{\infty}\frac{n^2+\sqrt{n}+2}{n^4-n^{4/3}+2},\qquad (2)\sum_{n=1}^{\infty}\frac{(-1)^n n}{n^2+2n},\qquad (3)\sum_{n=1}^{\infty}\frac{3n^2+5}{5n^2-3n+2},\qquad (4)\sum_{n=1}^{\infty}\frac{(-1)^n n}{3^n-2}.$$

Power series. Find the radius and interval of convergence for the following power series:

$$(5)\sum_{n=0}^{\infty}\frac{n}{(n^2+1)5^n}(x-3)^n,\qquad (6)\sum_{n=0}^{\infty}\frac{(n!)^2}{(3n)!}(x+2)^n.$$

Taylor series. Find Taylor series for the following functions:

$$(7)\ \frac{x}{3-x},\qquad (8)\ \sin(2x^2),\qquad (9)\ \frac{1}{\sqrt{1-x^2}},\qquad (10)\ \arcsin(x).$$

Integration. Evaluate the following integrals:

$$(11)\int \cos^5(x)\,dx,\qquad (12)\int\frac{1}{(1-4x^2)^{3/2}}\,dx,\qquad (13)\int x^3e^x\,dx,\qquad (14)\int\frac{4+x}{(1+2x)(3-x)}\,dx.$$

Miscellaneous.

(15) Is the function x^3-4x invertible?

(16) Compute $\lim\limits_{x\to\infty}\dfrac{1+\ln(x)+(\ln(x))^2}{\sqrt{x}}$.

(17) Compute the arc length of the curve $x=\frac{2}{3}(y^2+1)^{3/2}$ for $2\le y\le 4$.

(18) Find a (specific) solution to the initial value problem $y'/x-y=1$ if $y(0)=3$.

(19) Let $b_n=\dfrac{(3n)!}{(n+1)(3n-1)!}$. Compute $\lim\limits_{n\to\infty} b_n$, with justification.

(20) Compute $\displaystyle\sum_{n=1}^{\infty}\frac{(-4)^{n-1}}{5^n}$, $\displaystyle\sum_{n=1}^{\infty}\frac{-4^{n-1}}{5^n}$, $\displaystyle\sum_{n=1}^{\infty}\frac{4^{n-1}}{5^n}$, and $\displaystyle\sum_{n=1}^{\infty}\frac{(-5)^{n-1}}{4^n}$.